

# Joel Daniel Andersson

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## Research interests

My work centers on the theory of differential privacy, with a focus on problems in the *online* setting (continual observation). By drawing on techniques from linear algebra and streaming, I aim to improve accuracy-privacy trade-offs with low resource overhead. More recently, I have also been exploring space complexity for online problems outside the context of privacy.

## Education

- 2022-2025    Ph.D. in Computer Science, University of Copenhagen  
              Title: *On Private Continual Counting*  
              Advisor: *Rasmus Pagh*
- 2014-2020    M.Sc. in Engineering Physics, Lund University  
              GPA: 3.93 / 4.0  
              Relevant coursework: algorithms, optimization, linear algebra, real analysis, combinatorics
- 2017-2018    Exchange year. University of California, San Diego  
              Formally enrolled as a CS student. Studied algorithms, complexity and probability theory.

## Professional experience

- 2025-        Institute of Science and Technology Austria (ISTA) – Klosterneuburg, Austria  
              Postdoc in the Monika Henzinger Group  
              Conducting research on differentially private algorithms.
- 2020-2022    Ericsson AB – Lund, Sweden  
              Software Engineer  
              Implemented protocols in C for the 5G Physical Layer in base stations.
- 2019-2020    CERN – Geneva, Switzerland  
              Technical Student (salaried 14-month internship while writing my master's thesis)  
              Designed a framework in Python for first-order closed orbit analysis in synchrotrons.
- 2018        CERN – Geneva, Switzerland  
              openlab Summer Student  
              Systematized the comparison of beam tracking codes and fixed identified bugs.

2016, 2017 Qlik AB – Lund, Sweden  
Summer Software Engineering Intern  
Revamped a test framework and automated documentation generation.

## Grants, honours & awards

2017 Gull & Stellan Ljungberg Foundation Scholarship  
2014 Hvitfeldtska Trust Scholarship  
2014 Honorable mention in International Physics Olympiad  
2014 Fifth place in Wallenberg (Swedish national) Physics Price Competition

## Publications

Authors are listed in alphabetical order by default. First-authors who are first by contribution have a “\*” next to their name.

### PEER-REVIEWED PUBLICATIONS

- 2026
9. **Improved Accuracy for Private Continual Cardinality Estimation in Fully Dynamic Streams via Matrix Factorization**  
J. D. A., Pakak Jain, Satchit Sivakumar.  
*To appear at Symposium on Principles of Database Systems (PODS), 2026.*  
[\[arxiv\]](#)
  8. **Learning Rate Scheduling with Matrix Factorization for Private Training**  
Nikita Kalinin\*, J. D. A.  
*To appear at Symposium on Foundations of Responsible Computing (FORC), 2026.*  
[\[arXiv\]](#)
- 2025
7. **Private Lossless Multiple Release**  
J. D. A.\*, Lukas Retschmeier, Boel Nelson, Rasmus Pagh.  
*International Conference on Machine Learning (ICML), 2025.*  
[\[arXiv\]](#) [\[proceedings\]](#) [\[poster\]](#)
  6. **Count on Your Elders: Laplace vs Gaussian Noise**  
J. D. A., Rasmus Pagh, Teresa Anna Steiner, Sahel Torkamani.  
*Symposium on Foundations of Responsible Computing (FORC), 2025.*  
[\[arXiv\]](#) [\[proceedings\]](#)
  5. **Streaming Private Continual Counting via Binning**  
J. D. A., Rasmus Pagh.  
*IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2025.*  
[\[arXiv\]](#) [\[proceedings\]](#)
- 2024
4. **Continual Counting with Gradual Privacy Expiration**  
J. D. A., Monika Henzinger, Rasmus Pagh, Teresa Anna Steiner, Jalaj Upadhyay.  
*Conference on Neural Information Processing Systems (NeurIPS), 2024.*  
[\[arXiv\]](#) [\[proceedings\]](#) [\[poster + 5 min video\]](#)

- 2023 3. **A Smooth Binary Mechanism for Efficient Private Continual Observation**  
J. D. A., Rasmus Pagh.  
*Conference on Neural Information Processing Systems (NeurIPS)*, 2023.  
[arXiv] [proceedings] [poster + 5 min video]
- 2019 2. **SixTrack V and Runtime Environment**  
Riccardo De Maria\* et al.  
*International Journal of Modern Physics A* 34 (36), 2019.  
[article]
1. **SixTrack Version 5: Status and New Developments**  
Riccardo De Maria\* et al.  
*Journal of Physics: Conference Series* 1350 (1), 2019.  
[article]

#### NON-PEER-REVIEWED PUBLICATIONS

- 2025 · **On the Space Complexity of Online Convolution**  
J. D. A., Amir Yehudayoff.  
*Preprint under review.* [arxiv]
- 2020 · **Orbit Correction Studies on the HL-LHC Layout and Optics V1.5**  
J. D. A., Riccardo De Maria, Davide Gamba.  
*Technical report, CERN.* [link]
- **A Linear Framework for Orbit Correction in the HL-LHC**  
J. D. A. (advisors: Davide Gamba and Alexandros Sopasakis).  
*Master's thesis, Lund University.* [link]

#### Talks

- 2026 · Presented at the **ISTA TCS Seminar**, March 6, 2026. [event link]  
45 minute talk on *Improved Accuracy for Private Continual Cardinality Estimation in Fully Dynamic Streams via Matrix Factorization*.
- 2025 · Presenting a paper at **FORC 2025**, June 5, 2025.  
15 minute talk on *Count on Your Elders: Laplace vs Gaussian Noise*.
- Invited talk at the **Boston University Security Seminar**, April 2, 2025.  
45-minute talk on *Streaming Private Continual Counting via Binning*.
- Invited talk at the **Google Privacy Seminar** hosted by Thomas Steinke, February 12, 2025.  
45-minute talk on *Streaming Private Continual Counting via Binning*.
- 2023 · Talk at **ARCO 2023**, November 24, 2023.  
20-minute talk on *A Smooth Binary Mechanism for Efficient Private Continual Observation*.
- Invited talk at the **Google Privacy Seminar** hosted by Thomas Steinke, August 30, 2023.  
45-minute talk on *A Smooth Binary Mechanism for Efficient Private Continual Observation*.

#### Teaching

- 2022-2025 Teaching assistant for the first-year graduate class in algorithms: Advanced Algorithms and Data Structures (AADS) at the University of Copenhagen. Responsibilities included leading exercise classes, preparing the exam and grading hand-ins. [course description]

## Academic service

### REVIEWING

ICML'25, PPML@EurIPS'25, ICLR'26, SWAT'26, Journal of Artificial Intelligence Research.

### PROGRAM COMMITTEES

SaTML'26, TDP'26.

### MISCELLANEA

- 2023 · Organized the biannual ARCO (Algorithmic Research Cooperation around Øresund) event at University of Copenhagen together with Jacob Holm. [[homepage](#)]

## Media appearances

- 2024 · Media article(s) on *A Smooth Binary Mechanism for Efficient Private Continual Observation*. English [press release from KU](#), and Danish article in [Dagens Pharma](#).
- 2023 · Contributed to SCIENC KU's *Christmas Calendar*. Recorded a [1-minute video](#), sketching out a connection between Traveling Salesman and Santa's predicament around Christmas.

## Other skills

|           |                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------|
| Coding    | Python, C, C++, Java, MATLAB, L <sup>A</sup> T <sub>E</sub> X, bash, git                                               |
| Languages | Swedish ( <i>native</i> ), English ( <i>fluent</i> ), Danish ( <i>B2</i> ), French ( <i>B1</i> ), German ( <i>A1</i> ) |
| Running   | 5 kilometers (19:15), half-marathon (1:35:31), marathon ( <a href="#">3:32:57</a> )                                    |